QUIZ #2 Sec2

Is there a linear homog. ODE of order 2 for which

1. $f = \sin t$, $g = t$
2. $f = \sin t$, $g = t \sin t$

are solutions over $\mathbb{R}$?

For each case,
if YES, write down the ODE
if NO, tell why.

Solution

1 NO: $W[f,g](t) = \begin{vmatrix} t & \sin t \\ 1 & \cos t \end{vmatrix}$

$= t\cos t - \sin t$

is sometimes 0, sometimes $\neq 0$

This would contradict Abel's identity, if there were such ODE.

2 Similarly NO:

$$W = \begin{vmatrix} \sin t & t\sin t \\ \cos t & \sin t + t\cos t \end{vmatrix} = \sin^2 t$$

Hence in each case, there is no ODE which has the given pair as solutions over all $\mathbb{R}$.

Note: If you confine yourselves in an interval where W is never zero (i.e. in ②, choose $(0,\pi)$) then one can cook up an ODE which has the given pair as solns. For instance:

① $\begin{vmatrix} y & t & \sin t \\ y' & 1 & \cos t \\ y'' & 0 & -\sin t \end{vmatrix} = 0$ gives a linear ODE of order 2

$(t\cos t - \sin t)y'' + \cdots = 0$

This has t & $\sin t$ as solutions over any region where $t\cos t \neq \sin t$.

② Similarly

$\begin{vmatrix} y & \sin t & t\sin t \\ y' & \cos t & \sin t + t\cos t \\ y'' & -\sin t & \cdots \end{vmatrix} = 0$ is an ODE

that has $\sin t$ & $t\sin t$ as solns over any region where $\sin^2 t \neq 0$.